Insights & Executive summary

The provided dataset is **station master data** (station ID, name/address, latitude/longitude, banking/bonus flags, station key). It does **not** include any **revenue**, **cost**, **trip**, or **distance (km)** fields, so questions like **“highest/lowest revenue per km”** and **“revenue trends over years”** cannot be computed from this extract alone. What we *can* conclude today is primarily about **network coverage, data quality, and operational readiness by city/contract**.

What the current data can tell us (actionable now)

1) Coverage footprint by city/contract

- Use Station_Key (or “Contract Name” derived from it) to quantify **station counts by city/contract** and identify where the network is concentrated (e.g., multiple stations visible for **Mulhouse, Lyon, Seville**, etc. in the sample).
- This supports decisions like where to prioritize data feeds, QA checks, and operational monitoring.

2) Data quality and usability readiness

- Geo Valid (lat/long present and within bounds) enables a clear **map-ready coverage rate**; any invalid/missing coordinates block spatial analysis and routing-based KPIs.
- Is Test Station flags non-production stations; these should be excluded from performance reporting to avoid skew.

3) Product/service feature coverage

- Banking and Bonus allow a quick segmentation of stations by payment capability and bonus eligibility—useful for identifying **service parity gaps** by city/contract (e.g., “which cities have low banking-enabled station coverage?”).

Why the required revenue questions cannot be answered (and what’s needed)

To answer:

- **Which cities earned highest/lowest revenue per km?**
- **How have revenues changed over the years?**
- **Which cities are underperforming with low cost efficiency?**

You need at minimum (at city and date granularity):

1. **Revenue**: fare revenue (and ideally pass/subscription allocation rules) with a **date** field.
2. **Distance (km)**: either actual **trip kilometers** (preferred) or estimated distance from trip OD (origin/destination) coordinates.
3. **Cost**: operating cost by city/date (maintenance, rebalancing, staffing, capex allocation), or a proxy (service hours, truck rolls, etc.).
4. **Time series structure**: a transaction or daily summary table with City/Contract, Date, Revenue, Trips, Km, Cost.

Without those, any “revenue per km” or “cost efficiency” conclusion would be speculative.